ModelIt!

Student Activity Pack

How Does Your Phone Hear Your Voice?

The Invisible Science of Sound Waves

Grade 4 | NGSS 4-PS4-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think sound is? How does it get from one place to another?

2 Why do you think you can hear someone even when you cannot see them?

3 Draw what you think happens in the air between your mouth and someone's ear when you talk.

4 How do you think a phone knows when you are talking to it?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sound Wave Strength | Vibration Size | Volume Level

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Whisper

Sound Wave Strength set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Shout

Sound Wave Strength set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Normal Voice

Sound Wave Strength set to 50%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Distance from Source — How far away you are from the sound — sound gets quieter the farther it travels

Material Type — What the sound is traveling through — sound moves differently through air, water, and solids

Pitch — How high or low a sound is — this depends on how fast the vibrations happen

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When you talk to your phone, how does it turn the invisible vibrations in the air into something it can understand? Use evidence from your simulation to support your answers.

1 If sound is made of vibrations, what is vibrating when you hear music from a speaker?

2 Why is it impossible to hear sounds in outer space?

3 How is a loud shout different from a soft whisper in terms of what the air is doing?

4 What would happen if you tried to talk to someone on the Moon with no equipment?

5 How did your model help you understand something invisible like sound waves?

8 STEM Challenge: Build a Sound Wave Catcher

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YOUR ENGINEERING MISSION

Design a device that can detect and show sound waves using simple materials, proving that sound is made of real vibrations.

THE SCENARIO

A music studio needs a way to SHOW musicians how loud they are playing without using electronics. Your engineering team must design a simple device that makes sound waves visible so musicians can see how strong their sound is.

Engineering Design Thinking

1. WHAT MATERIAL WILL YOU USE TO CATCH SOUND VIBRATIONS (PLASTIC WRAP, PAPER, FOIL)?
2. HOW WILL YOU MAKE THE VIBRATIONS VISIBLE (SALT, RICE, GLITTER ON THE SURFACE)?
3. HOW WILL YOU TEST YOUR DEVICE WITH DIFFERENT VOLUMES?
4. HOW WILL YOU PROVE THAT LOUDER SOUNDS CAUSE BIGGER VIBRATIONS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sound Engineers and Audio Technicians use science to record, mix, and design sound for music, movies, and live events. They earn $45,000–$95,000/year.